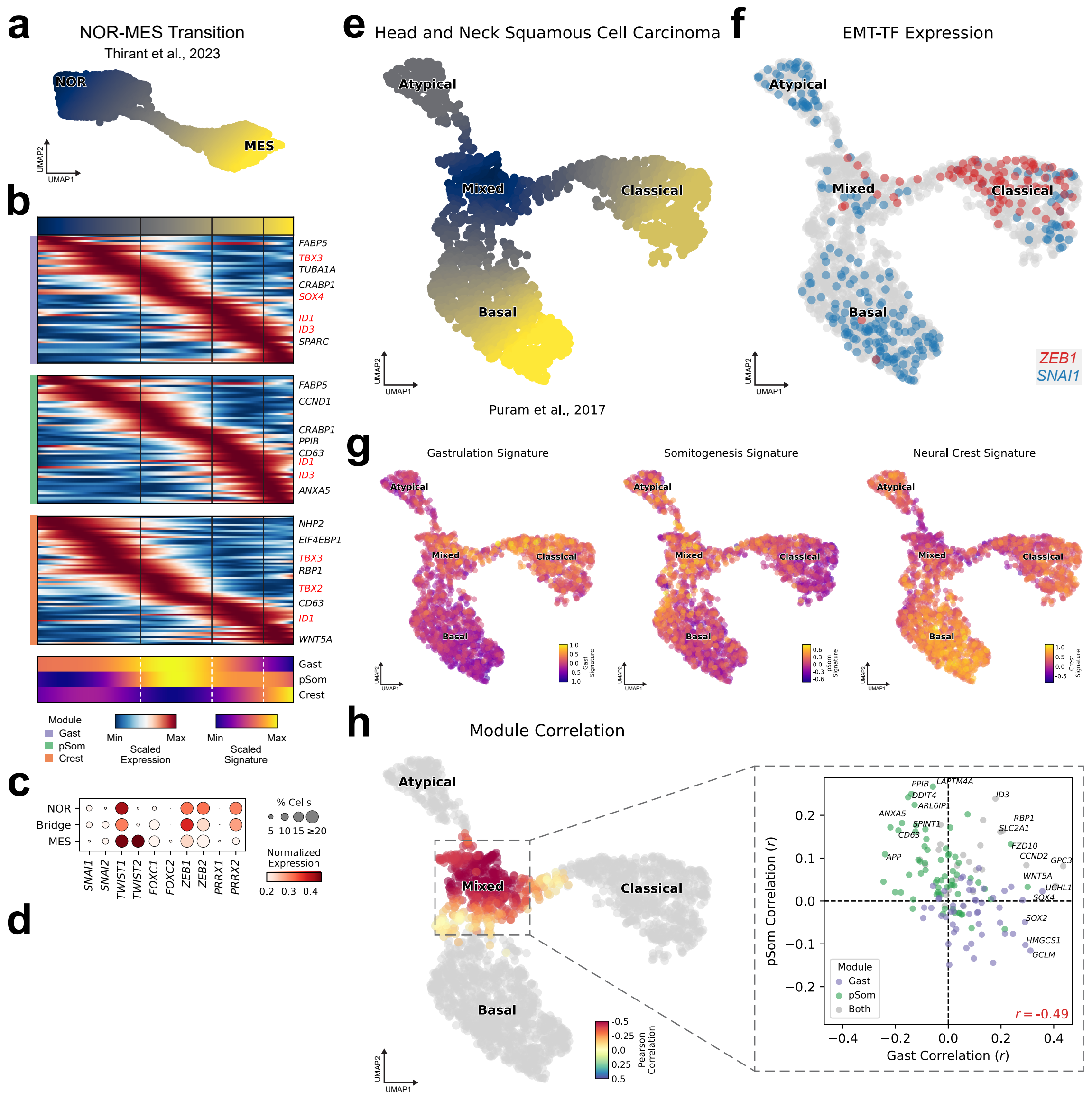

a
NOR-MES Transition
Thirant et al., 2023
NOR
MES
UMAP2
UMAP1
b
FABP5
TBX3
TUBA1A
CRABP1
SOX4
ID1
ID3
SPARC
FABP5
CCND1
CRABP1
PPIB
CD63
ID1
ID3
ANXA5
NHP2
EIF4EBP1
TBX3
RBP1
TBX2
CD63
ID1
WNT5A
Gast
pSom
Crest
Module
Gast
pSom
Crest
Min
Max
Scaled Expression
Min
Max
Scaled Signature
c
NOR
Bridge
MES
SNAI1
SNAI2
TWIST1
TWIST2
FOXC1
FOXC2
ZEB1
ZEB2
PRRX1
PRRX2
% Cells
5 10 15 ≥20
Normalized Expression
0.2 0.3 0.4
d
e
Head and Neck Squamous Cell Carcinoma
Atypical
Mixed
Classical
Basal
UMAP2
UMAP1
Puram et al., 2017
f
EMT-TF Expression
Atypical
Mixed
Classical
Basal
ZEB1
SNAI1
UMAP2
UMAP1
g
Gastrulation Signature
Somitogenesis Signature
Neural Crest Signature
Atypical
Mixed
Classical
Basal
1.0
0.5
0.0
-0.5
-1.0
Gast Signature
0.6
0.3
0.0
-0.3
-0.6
pSom Signature
1.0
0.5
0.0
-0.5
Crest Signature
UMAP2
UMAP1
h
Module Correlation
Atypical
Mixed
Classical
Basal
-0.5
-0.25
0.0
0.25
0.5
Pearson Correlation
UMAP2
UMAP1
PPIB
LAPTM4A
DDIT4
ID3
ARL6IP1
ANXA5
RBP1
SPINT1
SLC2A1
CD63
FZD10
APP
CCND2
GPC3
WNT5A
UCHL1
SOX4
SOX2
HMGCS1
GCLM
Module
Gast
pSom
Both
r = -0.49
pSom Correlation (r)
Gast Correlation (r)
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.2
0.0
0.2
0.4